

A Constant Beats Every Arm: Lessons from Re-analysing a Privacy-Harm Heuristics Evaluation

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Abstract

We re-analyse a small study that asked whether interpretable heuristics can estimate privacy harm, and how they compare with large language models (LLMs). Its recorded result was that injecting a static block of privacy rules into an LLM prompt beat a plain LLM, retrieval augmentation, and two hybrids on all four reported metrics. Using only the study’s committed artifacts and no model calls, we find that constant predictors beat every arm on three of those four metrics: predicting *no harm* for every case gives the highest exact-match ratio (0.70), and predicting all four of Solove’s harm groups for every case gives the highest Jaccard (0.175) and nDCG@5 (0.282). On the fourth metric, the only run with per-case outputs cannot be distinguished from a constant. We trace this to an empty-majority, LLM-generated gold set; to dry runs stored beside real results; and to training labels produced entirely by a keyword fallback. We distil nine lessons for anyone evaluating harm classifiers built on LLMs.

1 Introduction

Estimating the *harm* of a privacy incident, not only detecting that one occurred, is useful to regulators, auditors, and engineers deciding what to fix first. Legal scholarship offers structure for the task: Solove’s taxonomy organises privacy problems into information collection, information processing, information dissemination, and invasion [14], and Citron and Solove catalogue the injuries that follow [3]. A natural engineering question is whether cheap, interpretable heuristics can apply that structure at scale, and whether an LLM does better.

A 2025 master’s practicum asked exactly that, in two questions: *can we estimate privacy harm using heuristics, given facts parsed by a small language model*, and *how do non-language models compare with LLMs and with an LLM applying classical privacy frameworks?* Its public artifact records a headline comparison in which a “rules static” arm led every metric.

This paper is the study’s close-out. We re-examined every committed artifact offline and asked what the evidence can support. The answer is: much less than recorded, but something useful about how such evaluations fail. Our contributions are:

- a re-analysis showing that trivial baselines, scored with the study’s own metric code, beat every recorded arm on three of four metrics (§3);
- an audit of the evaluation’s provenance: its gold set, its dry

- runs, and where its training labels came from (§4);
 - nine lessons, each tied to the evidence behind it (§5).
- One script regenerates every re-analysis number from the public repository without network access (§7).

2 The original study

Pipeline. Public incident reports (regulator breach filings, securities filings, news, forums, Wikipedia) were crawled, labelled with Solove categories, and turned into features. Five interpretable model families were trained: decision tree, Explainable Boosting Machine [10, 11], Bayesian rule lists [9], a sparse linear model, and a Chow–Liu Bayes net [2]. Their exported rules fed “hybrid” arms. Facts were to be extracted by GLiNER2, a successor to GLiNER [15].

Evaluation. Six arms were scored on a 50-case gold set: a plain LLM baseline; *rules static* (a fixed block of rules injected into the prompt); *rules dynamic*; RAG; and two hybrids. All LLM arms were served by Gemini; the recorded run did not log a model version. The metrics were instance-averaged Jaccard, exact-match ratio (EMR, i.e. subset accuracy), micro-F1, and nDCG@5 over predicted harm groups [16]. The recorded table (Table 1, top) is a mean of five trials. A 2026 rerun pinned the model (gemini-2.5-flash) and repeated the rules-static arm once. The repository also added an *offline deterministic* arm that uses keyword rules with no LLM, so at least one arm reproduces byte-for-byte.

3 Constants beat the arms

The gold set is mostly empty. Of the 50 gold cases, 35 carry no harm label. After mapping to Solove’s four groups, the other 15 carry 35 labels (processing 13, invasion 11, collection 7, dissemination 4). An empty prediction is therefore correct 70% of the time, and EMR, which counts only exact set equality, mainly measures how often an arm predicts nothing.

Result. Table 1 adds four constant predictors. The empty predictor’s EMR (0.700) exceeds every arm’s. Predicting all four groups for every case beats every arm on Jaccard and nDCG@5. It also beats the offline keyword arm on every metric except EMR. Only one number in the recorded table, rules-static micro-F1 (0.391), exceeds every constant (best 0.306), and no per-case outputs of that run survive to test the gap. The

Arm	Jaccard	EMR	micro-F1	nDCG@5
<i>Recorded, Nov 2025 (model unpinned, 5 trials)</i>				
Rules static	0.068	0.667	0.391	0.090
LLM baseline	0.056	0.620	0.285	0.072
Rules dynamic	0.055	0.593	0.281	0.073
Hybrid (det. first)	0.053	0.593	0.275	0.068
RAG	0.047	0.580	0.242	0.061
Hybrid (LLM first)	0.036	0.600	0.195	0.049
<i>Later runs</i>				
Rules static, pinned (2026)	0.105	0.360	0.336	0.176
Offline keyword rules	0.081	0.080	0.222	0.162
<i>Constant predictors (this paper)</i>				
Always $\emptyset$	0.000	0.700	0.000	0.000
Always {processing}	0.127	0.040	0.306	0.154
Always {invasion}	0.097	0.020	0.259	0.122
Always all four groups	0.175	0.040	0.298	0.282

Table 1: All rows score the same 50-case gold set. Constant predictors are scored with the repository’s own metric code, which reproduces the pinned 2026 run exactly. Bold marks the best value in each column.

pinned rerun does keep per-case outputs. Its micro-F1 (0.336; 95% bootstrap CI [0.199, 0.451], 2,000 resamples [5]) minus the constant {processing} predictor’s, paired over cases, has CI [−0.080, 0.135]: no detectable difference.

Where the arm’s EMR came from. In the pinned rerun, the arm predicted the empty set on 18 of the 35 empty-gold cases and exactly matched none of the 15 non-empty cases. Its predictions also used eight labels outside the gold label space (root-cause terms such as `lack_of_consent`), and each one is a guaranteed false positive under set-based scoring.

Severity. The rerun also scored 1–5 harm severity. It reached accuracy 0.50 (quadratic-weighted $\kappa = 0.33$). Because 39 of the 50 gold severities are 1, always predicting 1 reaches 0.78.

Drift. The pinned rerun of the same arm on the same cases moved EMR from 0.667 to 0.360 and roughly doubled Jaccard and nDCG@5. The newer model predicts non-empty sets far more often; that costs it on the empty-majority set and helps it on overlap metrics. Hosted models change under a stable name [1]; with no model version recorded, the 2025 table cannot be reproduced by construction.

4 How the evaluation misled

An LLM-generated, mostly off-topic gold set. The experiment summary records that the 50-case gold set was generated with Gemini 2.5 Pro, and the LLM arms were served by Gemini. Scoring a model family against labels written by the same family invites self-preference [12, 17], and here nothing measures it. The cases are crawled pages, not incident reports: 47 of the 50 are general Wikipedia articles (several countries’ nationality

laws, a law school, a flowering shrub, Moore’s law, underwater diving). The 15 cases labelled with harm include the Railway Regulation Act 1844 and an article on data warehousing. None has the `harms` field that the scorer reads first, so the scorer silently fell back to `root_causes`, a field its own comment says holds technical causes, not Solove harms. A separate 96-case set, with labels drafted by `gemini-2.5-flash`, was marked as reviewed through a web review tool in July 2026 (78 confirmed, 5 corrected, 2 annotated, 11 without a note). The record does not say whether the reviewer was a person, and no recorded live comparison used this set.

Dry runs stored as experiments. The repository keeps 14 “comprehensive” run directories, the largest spanning 320 configurations with confidence intervals, Bonferroni-corrected pairwise tests, and Cliff’s δ . Seven are flagged `dry_run` in their metadata. The five that report per-arm metrics are all dry runs, and within each the mean micro-F1 is identical to 12 decimal places across every arm in four of them, with two distinct values in the fifth. The seven live runs committed no per-arm metrics. The statistics machinery ran on placeholder outputs, correctly found “no significant difference,” and produced tables that look like results.

Heuristic labels in, heuristic rules out. The labeller was designed to assign a Solove category with GLiNER2, then fall back to keyword scoring, then to an LLM. In the released 33,472-row corpus, every label’s recorded source is the keyword fallback, and 61.4% of labels are unknown. The interpretable models were trained on seven features: a log count of affected individuals and six keyword flags. They therefore learned to approximate one keyword heuristic from another, which cannot answer the study’s first question. They did so poorly. Macro-F1 was 0.14 (decision tree), 0.10 (EBM), 0.20 (sparse linear), and 0.04 (Bayes net). The Bayes net predicted a single class for every test row; its accuracy, 0.524, is therefore that class’s share of its test split. The other three accuracies (0.58–0.62) sit at most 0.10 above it, although the repository does not record whether the splits were identical. Of the 16 per-category rule-list classifiers, one trained successfully. The decision tree’s 20 exported “heuristics” each have support 0.000 and precision at most 0.004, and they name classes by index rather than label.

Unresolvable provenance. The trained models record a SHA-256 for their training corpus. It identifies a pre-scrub file that no archived copy we could locate matches; the public release is the scrubbed derivative.

The dataset enacted the harm it studied. A pre-release audit found that the social-media portions of the corpus republished verbatim, re-identifiable posts in which private individuals described legal, medical, and financial trouble. That is the secondary-use and aggregation harm the project set out to measure. A tested scrub removed handles, bodies, and permalinks before release.

5 Lessons

- L1 Report constant baselines with the same scorer.** Include empty, modal, and all-labels predictors for sets, and the modal score for ordinal targets. A metric on which an arm does not beat every constant supports no claim (§3).
- L2 Treat empty-majority EMR as an emptiness detector.** Report the gold empty share and score the non-empty subset separately.
- L3 Keep the gold set independent of the systems under test.** Do not generate it with the model family you evaluate, and have a person review it before the headline run, not after.
- L4 Make scorer fallbacks loud.** A missing field that silently switches the target from harms to causes changed what was measured.
- L5 Make placeholder runs unable to pass for results.** Refuse to aggregate or test when outputs do not vary by arm, and keep dry-run output out of results directories [8].
- L6 Measure the label pipeline you ran, not the one you designed.** Log the share of labels from each source; a 100% fallback share means the models learn the fallback [13].
- L7 Pin model versions and keep per-case outputs.** Summaries cannot be re-tested, and unversioned hosted models turn a table into a dated anecdote.
- L8 Name arms by mechanism, and include a model-free arm.** “Rules static” still used an LLM to produce every label.
- L9 Scrub at collection, not at release.** Privacy datasets can reproduce the harm they study.

These lessons echo known failure modes: annotation artifacts that let trivial models score well [7], shortcut learning [6], leakage between labels and features [8], and benchmark rankings that are sensitive to design choices [4]. What is specific to LLM-era harm classification is how cheaply an LLM can produce each ingredient (gold labels, rules, predictions), and so how easily one model family ends up on both sides of the comparison.

6 Limitations of this re-analysis

We made no model calls. We therefore cannot say how the arms would score on the reviewed 96-case set. The constant baselines use the current scorer; the 2025 table was produced by an earlier revision whose label mapping was not recorded. The empty-predictor EMR depends only on which gold cases are empty, so that comparison is robust to the mapping; the Jaccard and nDCG comparisons assume the mapping did not change. The model analysis relies on the recorded metrics files and the scrubbed public corpus, because the training corpus could not be matched.

7 Conclusion

The study’s headline, that a static rule block beats LLM baselines at estimating privacy harm, does not survive a constant predictor. Its first question, whether heuristics can estimate

harm, remains open, because the labels it learned from were themselves heuristics. What the study does offer is a compact, reproducible case of how an evaluation can pass every internal check (CI, confidence intervals, corrected tests) and still measure almost nothing. We hope the nine lessons save the next project the same year.

Artifact availability

The repository, including gold sets, recorded runs, trained models, and a scrubbed corpus (release v1.0.0), is public at <https://github.com/Savage-Fred/privacy-harm-heuristics>. `paper/reanalysis.py` regenerates `paper/reanalysis.json`, the source of every re-analysis number here. It makes no network calls and asserts that its scoring path reproduces the recorded 2026 rerun exactly.

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